Numerical Optimization

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Numerical Optimization

Objective function
- Value we seek to minimize or maximize

Decision variables
- Variables we can change

Constraints
- Constraints on model

Numerical Optimization

$$\max_{\text{Decision Variables}} \quad \text{Objective Function}$$

$$\text{s.t.} \quad \text{Constraints}$$

Application: Portfolio Theory

- Given a menu of assets, what do we invest in and how much?
- n assets
  - R_{it} is the return on asset i at time t
- Choose investment weights
  - w_i weight in asset i
- Given investment weights, portfolio return at time t is

$$R_t^P = w_1 \cdot R_{1t} + \cdots + w_n \cdot R_{nt}$$

Objective Function

Maximize Sharpe ratio

- Reward-to-Risk

$$\max \frac{E\left[R^P\right] - R^f}{\sigma_P}$$

Portfolio Statistics

Given historical data

- $E\left[R^P\right] = \text{average}\left(R_t^P\right)$
- $\sigma_P = \text{stdev.s}\left(R_t^P\right)$

- Analytical formulas for two assets
  - $E\left[R^P\right] = w_1 E[R_1] + w_2 E[R_2]$
  - $\sigma_P = \sqrt{\sigma_P^2}$
  - $\sigma_P^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2 w_1 w_2 \sigma_{12}$
  - $\sigma_{12} = \rho_{12} \sigma_1 \sigma_2$

Decision Variables

Choose investment weights

- w_i weight in asset i

Constraints

Sum of weights totals 100%

$$\sum_{i=1}^{n} w_i = 1$$

No short selling

$$w_i \geq 0$$

Constraints

- Constraints must be expressed as inequalities or equations
- Look for key words in word statements of constraints

cannot exceed	$\leq$
at least	$\geq$
exactly	$=$

Solutions

Unbounded Solution

- Objective can be increased without bound to infinity for a maximization problem or decreased to negative infinity for a minimization problem

Infeasible Solution

- Solution that violates at least one constraint
- Model is infeasible if all solutions are infeasible

Feasible Solution

- Any solution that satisfies all of the constraints

Optimal Solution

- Feasible solution achieves best objective function

Unique Optimal Solution

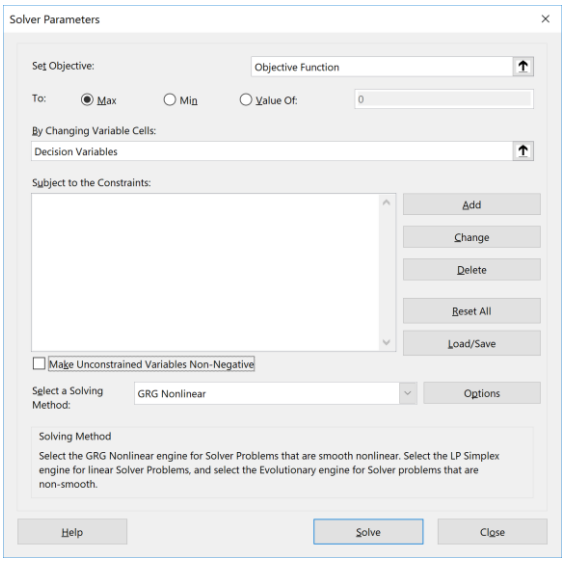
- Exactly one optimal solution

Constraints

- Binding constraint
  - Constraint holds with equality at optimal solution
- Slack constraint
  - Constraint is not binding

Solving Optimization Models

Solver is an Excel add-in for solving optimization problems



Sensitivity Analysis

What happens when we change the:

- Objective function?
- Constraints?
- Parameters?
- How sensitive are our optimal decision variables?

- Portfolio theory we will explore changing:
  - Objective function: alternative risk-reward preferences
  - Constraints: investment limits
  - Parameters: correlation, number of assets